

1 Solution — GIAM 4.2

Problem 1

1.1 Problem

Insert either $\in$ or $\subseteq$ in the blanks in the following sentences (in order to produce true sentences).

- i) 1 ____ $\{3, 2, 1, \{a, b\}\}$ iii) $\{a, b\}$ ____ $\{3, 2, 1, \{a, b\}\}$
ii) $\{a\}$ ____ $\{a, \{a, b\}\}$ iv) $\{\{a, b\}\}$ ____ $\{a, \{a, b\}\}$

1.2 Answer

i) $\in$	ii) $\subseteq$	iii) $\in$	iv) $\subseteq$
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1.3 The Two Tests

Everything turns on a single distinction:

$$\begin{array}{l|l} L \in R & \text{is } L \text{ itself one of the things listed inside } R? \\ L \subseteq R & \text{is every } \textit{element of } L \text{ one of the things listed inside } R? \end{array}$$

The first treats L as a single object; the second opens L up and looks at its contents. Note also that $L \subseteq R$ only makes sense when L is a set.

1.4 Case by Case

i) $1 \in \{3, 2, 1, \{a, b\}\}$. The right-hand side lists four things: 3, 2, 1 and $\{a, b\}$. The number 1 is the third of them, so $\in$ holds. Note $\subseteq$ could not apply here in any case — 1 is not a set, so it has no elements to compare.

*ii) $\{a\} \subseteq \{a, \{a, b\}\}$. The right-hand side lists two things: a and $\{a, b\}$. The set $\{a\}$ is not one of them, so $\in$ fails. But the only element of $\{a\}$ is a , which **is** listed — so every element of the left side is in the right side, and $\subseteq$ holds.*

iii) $\{a, b\} \in \{3, 2, 1, \{a, b\}\}$. Here $\{a, b\}$ appears literally as the fourth item in the list, so $\in$ holds. And $\subseteq$ fails: its elements a and b are not among $3, 2, 1, \{a, b\}$.

iv) $\{\{a, b\}\} \subseteq \{a, \{a, b\}\}$. The left side is a set with one element, namely $\{a, b\}$. That element is listed on the right, so $\subseteq$ holds. But $\{\{a, b\}\}$ itself is not one of the two items listed, so $\in$ fails.

All eight membership and subset tests were evaluated directly, confirming that in each of the four blanks **exactly one** of the two symbols works.

1.5 The Picture

THE TWO TESTS

$L \in R$ — is L itself one of the things listed inside R ?

$L \subseteq R$ — is every element of L one of the things listed inside R ?

The first asks about L as a single object; the second opens L up and asks about its contents.

THE FOUR BLANKS

i)

$$1 \in \{3, 2, 1, \{a, b\}\}$$

1 is one of the four things listed.
It is not a set, so $\subseteq$ cannot even apply.

iii)

$$\{a, b\} \in \{3, 2, 1, \{a, b\}\}$$

$\{a, b\}$ is listed as the fourth element.
Not $\subseteq$: a and b are not themselves listed.

ii)

$$\{a\} \subseteq \{a, \{a, b\}\}$$

Its only element a is in the right side.
Not $\in$: the elements are a and $\{a, b\}$, not $\{a\}$.

iv)

$$\{\{a, b\}\} \subseteq \{a, \{a, b\}\}$$

Its only element $\{a, b\}$ is in the right side.
Not $\in$: $\{\{a, b\}\}$ itself is not listed.

BOTH TESTS RUN ON ALL FOUR — exactly one passes each time

	left side	right side	$\in$?	$\subseteq$?
i)	1	$\{3, 2, 1, \{a, b\}\}$	yes	no
ii)	$\{a\}$	$\{a, \{a, b\}\}$	no	yes
iii)	$\{a, b\}$	$\{3, 2, 1, \{a, b\}\}$	yes	no
iv)	$\{\{a, b\}\}$	$\{a, \{a, b\}\}$	no	yes

THE PATTERN IN THE FOUR ANSWERS

Compare (iii) and (iv). The right sides differ, but so does the *number of braces* on the left:

$$\{a, b\} \in \{3, 2, 1, \{a, b\}\}$$

$$\{\{a, b\}\} \subseteq \{a, \{a, b\}\}$$

Wrapping an object in one more pair of braces turns an $\in$ statement into a $\subseteq$ statement, because the extra layer is exactly what $\subseteq$ strips off before comparing.

All eight membership and subset tests were evaluated directly.

Figure 1.1: A diagram in four parts. The first states the two tests: L is an element of R asks whether L itself is one of the things listed inside R ; L is a subset of R asks whether every element of L is one of the things listed inside R . The first treats L as a single object, the second opens L up and asks about its contents. The centre shows four cards, one per blank. Case i, in green: 1 is an element of the set 3, 2, 1, and the set a b — because 1 is one of the four things listed, and subset cannot even apply since 1 is not a set. Case iii, also green: the set a b is an element of the set 3, 2, 1, and the set a b — because it is listed as the fourth element; not a subset, since a and b are not themselves listed. Case ii, in red: the set containing a is a subset of the set containing a and the set a b — because its only element a is in the right side; not an element, since the elements listed are a and the set a b , not the set containing a . Case iv, also red: the set containing the set a b is a subset of the set containing a and the set a b — because its only element, the set a b , is in the right side; not an element, since the set containing the set a b is not itself listed. A table then runs both tests on all four cases, showing element yes and subset no for cases i and iii, and element no and subset yes for cases ii and iv,

so exactly one passes each time. A final band describes the pattern in the four answers: comparing cases iii and iv, the right sides differ but so does the number of braces on the left. The set $a\ b$ is an element of the set $3, 2, 1$, and the set $a\ b$; while the set containing the set $a\ b$ is a subset of the set containing a and the set $a\ b$. Wrapping an object in one more pair of braces turns a membership statement into a subset statement, because the extra layer is exactly what subset strips off before comparing. All eight membership and subset tests were evaluated directly.

1.6 Remark — One Extra Pair of Braces Flips the Symbol

Parts (iii) and (iv) are the heart of the exercise. Strip away the different right-hand sides and look at what changed on the left:

$$\{a, b\} \in R \quad \text{versus} \quad \{\{a, b\}\} \subseteq R'.$$

Adding one pair of braces converts an $\in$ statement into a $\subseteq$ statement. That is not a coincidence — it is a general fact:

$$x \in R \quad \Longleftrightarrow \quad \{x\} \subseteq R.$$

The reason is immediate from the definitions. Saying $\{x\} \subseteq R$ means every element of $\{x\}$ lies in R , and $\{x\}$ has exactly one element, namely x . So the two statements say the same thing in different packaging.

This is the cleanest way to keep the symbols straight: $\subseteq$ **takes one layer of braces off the left-hand side before comparing, and $\in$ does not.**

1.7 Remark — Why $\subseteq$ Fails in (iii), and $\in$ in (ii)

The two failures are instructive in opposite ways.

In (iii), $\{a, b\} \subseteq \{3, 2, 1, \{a, b\}\}$ is false because the test asks about a and b individually, and neither is listed on the right. The right-hand side contains the *box* $\{a, b\}$ but not its *contents*. This is exactly the failure of transitivity noted in 4.1 Problem 3:

$$a \in \{a, b\} \in R \quad \text{does not give} \quad a \in R.$$

In (ii), $\{a\} \in \{a, \{a, b\}\}$ is false for the mirror-image reason: the right side contains a but not the box $\{a\}$. One might paraphrase: (iii) has the box but wants the contents, (ii) has the contents but wants the box.

1.8 Remark — Both Symbols Can Hold at Once, Though Not Here

In all four blanks exactly one symbol works, which makes the exercise clean. That is not forced in general — there are sets for which **both** relations hold:

$$\{\{a\}\} \in \{\{a\}, \{\{a\}\}\} \quad \text{and} \quad \{\{a\}\} \subseteq \{\{a\}, \{\{a\}\}\},$$

since the right side lists $\{\{a\}\}$ (giving $\in$) and also lists $\{a\}$, the sole element of the left side (giving $\subseteq$).

The simplest example of the phenomenon is $\emptyset$ and $\{\emptyset\}$, from 4.1 Problem 1:

$$\emptyset \in \{\emptyset\} \quad \text{and} \quad \emptyset \subseteq \{\emptyset\},$$

both true. So the exercise's neat one-symbol-each pattern is a property of the particular sets chosen, not a law. It is worth checking both tests rather than stopping at the first that works.

1.9 Remark — A Checklist for Problems of This Kind

The reliable procedure is mechanical.

1. **List the elements of the right-hand side explicitly.** For $\{3, 2, 1, \{a, b\}\}$ they are $3, 2, 1, \{a, b\}$ — four things, one of which happens to be a set.
2. **Ask whether the whole left-hand side appears in that list.** If yes, $\in$.
3. **If the left-hand side is a set, list its elements and ask whether each appears.** If yes, $\subseteq$.
4. **If the left-hand side is not a set, $\subseteq$ is not even a candidate** — as in part (i).

Steps 2 and 3 are different questions, and running both is the only way to be sure. The single most common error at this stage is to run step 3 when the problem wants step 2, which is precisely what part (iii) is testing.